Structural Circularity and Non-Self-Sufficiency in Formal Theories

A Formal Meta-Theoretical Result

Abstract

We prove that no non-trivial formal theory can establish the validity of its own inference rules without invoking either (i) external meta-theoretical assumptions or (ii) structural circularity. This result is established within a standard model-theoretic framework and does not rely on interpretative or ontological extensions.

1. Preliminaries

Definition 1 (Formal Theory)

A formal theory is a tuple:

$$T = (\mathcal{L}, A, R)$$

where:

- $\mathcal{L}$ is a formal language
- $A \subseteq \mathcal{L}$ is a set of axioms
- R is a set of inference rules

Definition 2 (Derivability)

A formula $\varphi \in \mathcal{L}$ is derivable in T, written:

$$T \vdash \varphi$$

if there exists a finite sequence of formulas ending in φ, each of which is:

- an axiom, or
- obtained from previous formulas by a rule in R

Definition 3 (Model and Semantics)

A model is a structure M such that:

$$M \models \varphi$$

is defined via a valuation function v assigning truth values to formulas of $\mathcal{L}$.

Definition 4 (Soundness)

A theory T is sound with respect to a class of models $\mathcal{M}$ if:

$$T \vdash \varphi \;\Rightarrow\; \forall M \in \mathcal{M},\; M \models \varphi$$

Definition 5 (Internal Rule Validation Statement)

Let $R = \{r_i\}$. A theory T internally validates its rules if, for each r_i, there exists a formula $\mathrm{Valid}(r_i) \in \mathcal{L}$ such that:

$$T \vdash \mathrm{Valid}(r_i)$$

and $\mathrm{Valid}(r_i)$ expresses that r_i preserves truth under all valuations.

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2. Problem Statement

Can a theory T establish the validity of its own inference rules using only its internal resources?

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3. Main Result

Theorem 1 (Non-Self-Sufficiency of Rule Validation)

Let $T = (\mathcal{L}, A, R)$ be a non-trivial formal theory. Then:

$$\text{If } T \vdash \mathrm{Valid}(R), \text{ then either:}$$

1. The proof relies on semantic notions not definable within T, or
2. The proof is circular, in that it presupposes the correctness of R

Proof

Assume that T proves $\mathrm{Valid}(R)$ using only its internal resources.

A derivation in T is constructed using rules in R. Therefore:

- Any proof of $\mathrm{Valid}(R)$ necessarily uses rules from R

Let $r \in R$. The derivation of $\mathrm{Valid}(r)$ must include at least one application of some rule r' $\in R$.

Thus, the derivation presupposes the correctness of r', and hence of R.

Therefore, the argument is circular unless:

- the proof of $\mathrm{Valid}(R)$ does not rely on R

But any derivation in T is defined in terms of R. Hence, avoiding the use of R requires stepping outside the theory (i.e., into a meta-theory).

Therefore:

- Either the validation is circular, or
- it depends on external semantics

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4. Corollary

Corollary 1 (Meta-Theoretical Dependence)

There is no non-trivial formal theory that can establish its own soundness without invoking a meta-theoretical framework.

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5. Discussion

This result is consistent with classical limitations in logic:

- Soundness proofs are typically carried out in a meta-theory
- Internal validation leads to circularity

The theorem does not depend on:

- specific logical systems
- particular semantics
- completeness assumptions

It is purely structural.

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6. Scope and Limitations

This result applies to:

- classical first-order logic
- higher-order systems
- formal deductive systems in general

It does not address:

- degrees of partial internal justification
- probabilistic or non-classical logics explicitly

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7. Conclusion

We have shown that:

> *A formal theory cannot fully justify its own inference rules without either circular reasoning or external semantic grounding.*

This establishes a structural limitation on formal systems independent of specific interpretations.

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8. Minimal Formal Insight

$$\text{Internal validation} \Rightarrow \text{Circularity} \vee \text{External semantics}$$